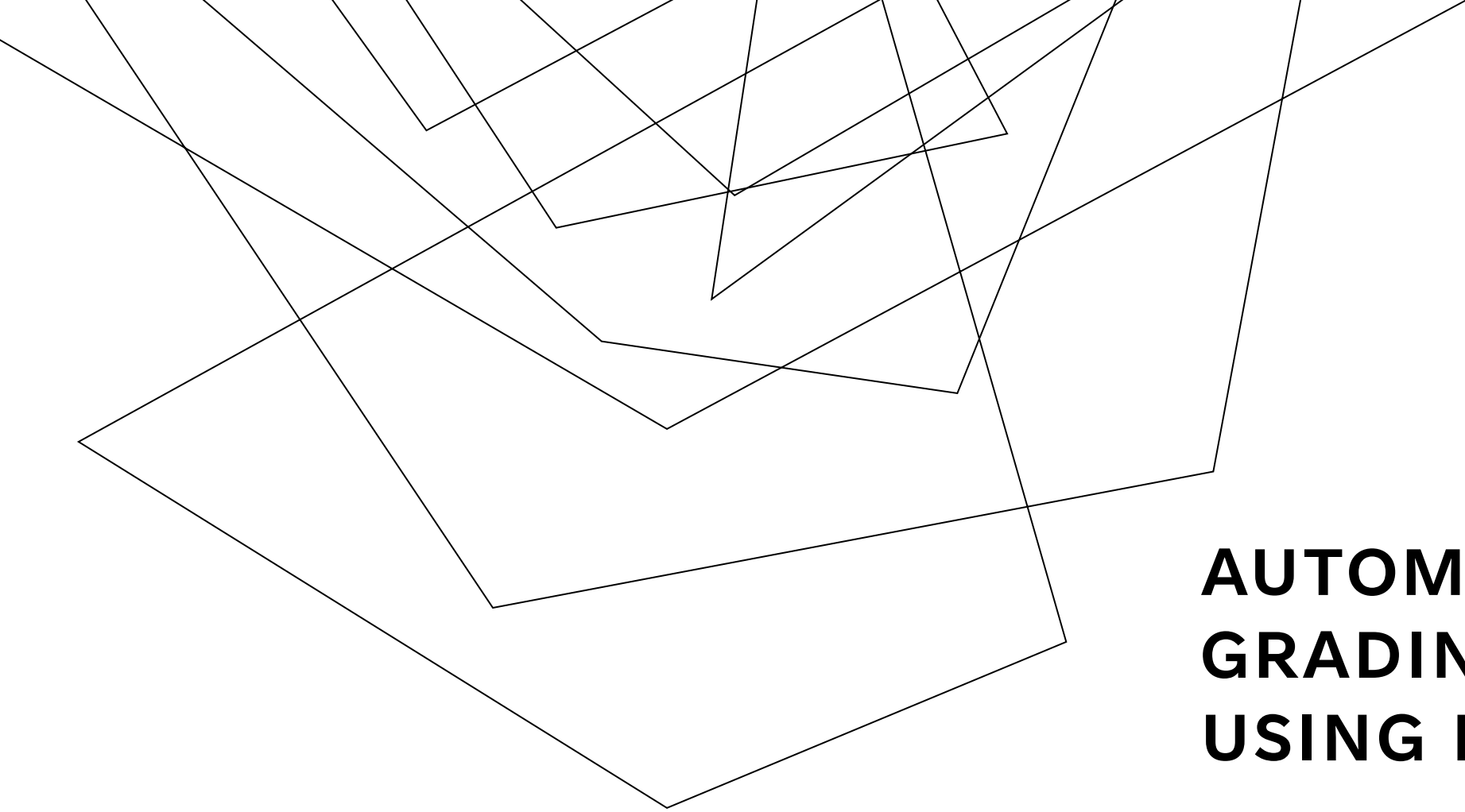


A series of thin, black, overlapping lines forming various geometric shapes like triangles and polygons, creating a complex, abstract pattern in the upper left portion of the slide.

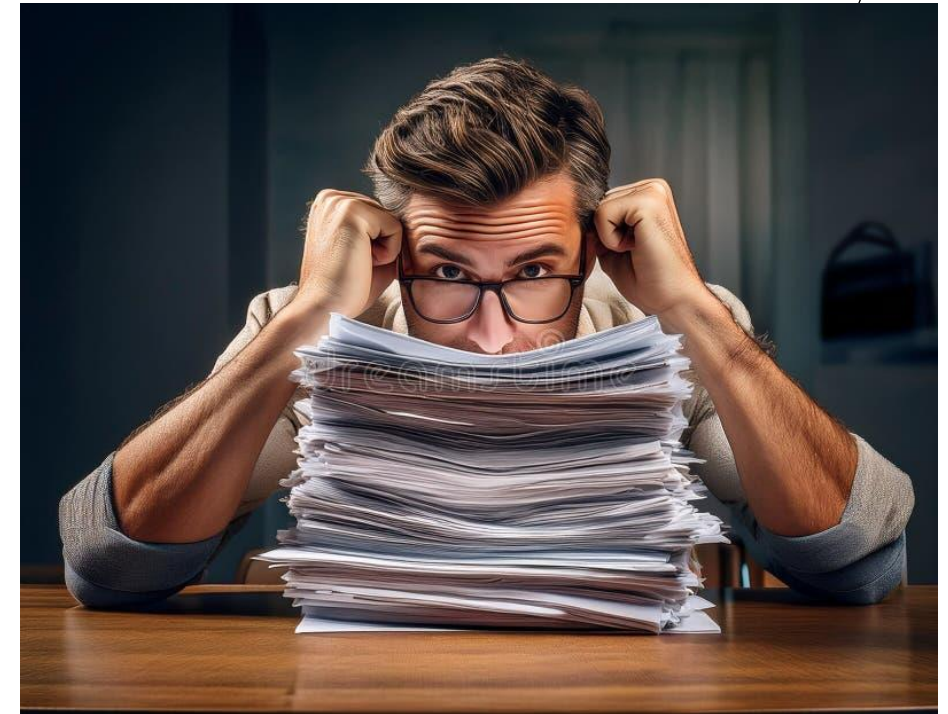
AUTOMATED GRADING SYSTEM USING MNIST

**TARGET AUDIENCE : PRIMARY & MIDDLE
SCHOOL MATH TEACHERS AND TUTORS IN
AFTER-SCHOOL PROGRAMS & LEARNING
CENTERS.**

BY – POORVI RAUT

INTRODUCTION :- THE GRADING BOTTLENECK

- Manually grading math worksheets is a time-consuming and resource-intensive task for teachers. This traditional approach can also lead to inconsistencies in evaluation and delays in providing students with essential feedback. These factors ultimately impact the efficiency of the educational process.
- Hence, an automated grading system is essential to assist educators by streamlining the evaluation process, ensuring accuracy, and providing timely feedback to students.

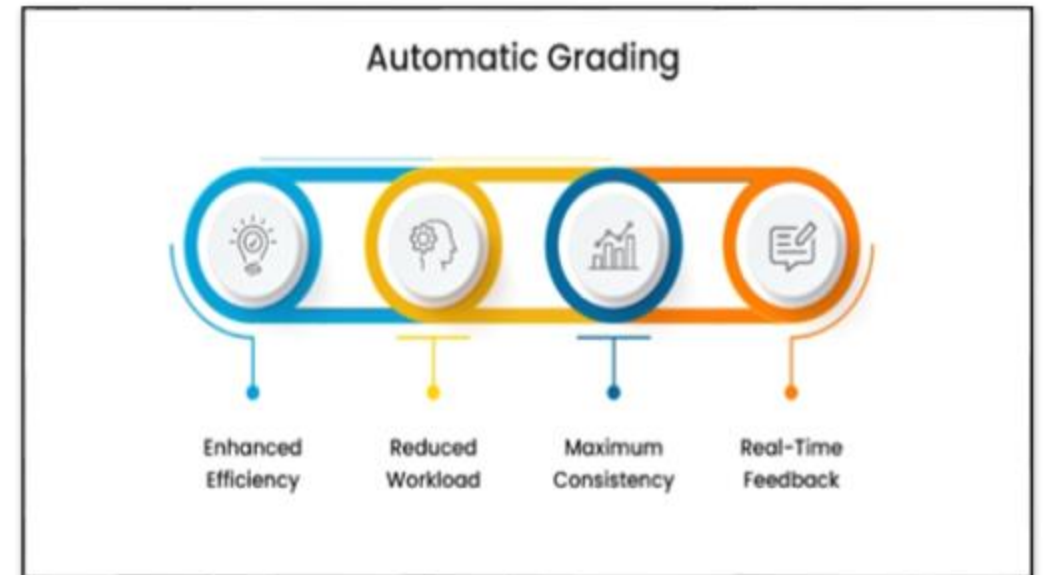


OUR SOLUTION: AUTOMATED GRADING SYSTEM

This system automates the grading of simple arithmetic problems in primary and middle school math worksheets by recognizing handwritten numbers and comparing them with correct answers.

Process:

- System recognizes handwritten digits in student's answers.
- Compares recognized digits with the answer key.
- Provides instant feedback and reduces teacher workload.



MNIST DATA-SET OVERVIEW

- A widely used dataset of handwritten digits.
- Contains 60,000 training images and 10,000 testing images.
- Each image is a 28x28 pixel grayscale representation of a digit (0-9).

•**Training Data:** 60,000 images (rows) x 787 columns (including 'Unnamed: 0', 'index', 'label' and 784 pixel values)

•**Testing Data:** 10,000 images (rows) x 785 columns (including 'Unnamed: 0', 'index', 'label' and 784 pixel values)

•Grayscale images: Pixel values range from 0 (black) to 255 (white).

•28x28 pixels = 784 features per image.

•Each image has a corresponding label (the actual digit it represents).

Train_data.csv

	Unnamed: 0	index	labels	0	1	2	3	4	5	6	...	774	775	776	777	778	779	780	781	782	783
0	0	0	5	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
1	1	1	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
2	2	2	4	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
3	3	3	1	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
4	4	4	9	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
59995	59995	59995	8	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
59996	59996	59996	3	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
59997	59997	59997	5	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
59998	59998	59998	6	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
59999	59999	59999	8	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0

60000 rows x 787 columns

Test_data.csv

	Unnamed: 0	index	labels	0	1	2	3	4	5	6	...	774	775	776	777	778	779	780	781	782	783
0	0	0	7	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
1	1	1	2	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
2	2	2	1	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
3	3	3	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
4	4	4	4	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
9995	9995	9995	2	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
9996	9996	9996	3	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
9997	9997	9997	4	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
9998	9998	9998	5	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
9999	9999	9999	6	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0

10000 rows x 787 columns

DATA PREPARATION

- Loaded MNIST data, removed index/unnamed columns, and checked for missing values.
- The training and test sets were split into features (X, X_test) and labels (y, y_test), then their shapes were determined.

```
[ ] train_data.info()
```

```
>>> <class 'pandas.core.frame.DataFrame'>
RangeIndex: 60000 entries, 0 to 59999
Columns: 787 entries, Unnamed: 0 to 783
dtypes: int64(787)
memory usage: 360.3 MB
```

```
[ ] train_data.describe()
```

```
>>>
```

	labels	0	1	2	3	4	5	6	7	8	...	7
count	60000.000000	60000.0	60000.0	60000.0	60000.0	60000.0	60000.0	60000.0	60000.0	60000.0	...	60000.0000
mean	4.453933	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.2004
std	2.889270	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	6.0424
min	0.000000	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0000
25%	2.000000	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0000
50%	4.000000	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0000
75%	7.000000	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0000
max	9.000000	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	254.0000

8 rows x 785 columns

```
>>>
```

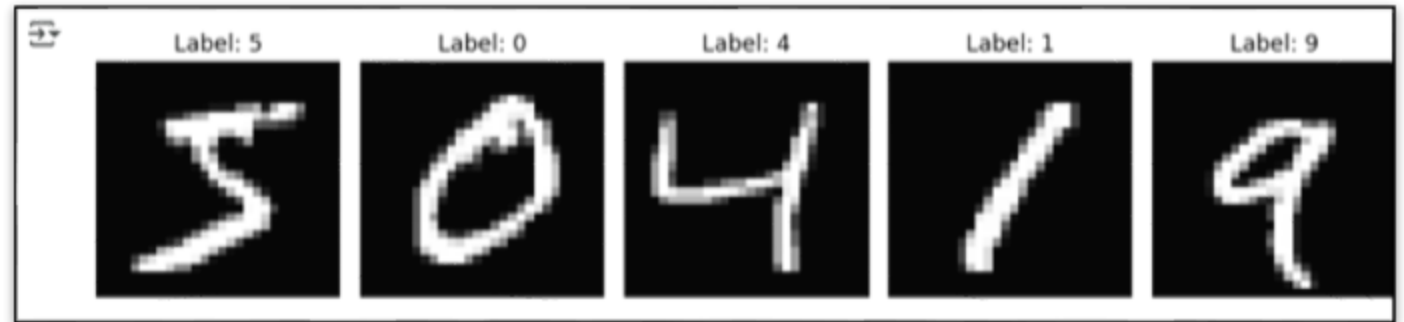
	labels	0	1	2	3	4	5	6	7	8	...	774	775	776	777	778	779	780	781	782	783
0	5	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
2	4	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
3	1	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
4	9	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0

```
>>> # Handling missing values
print("Missing values:", train_data.isnull().sum().sum())
```

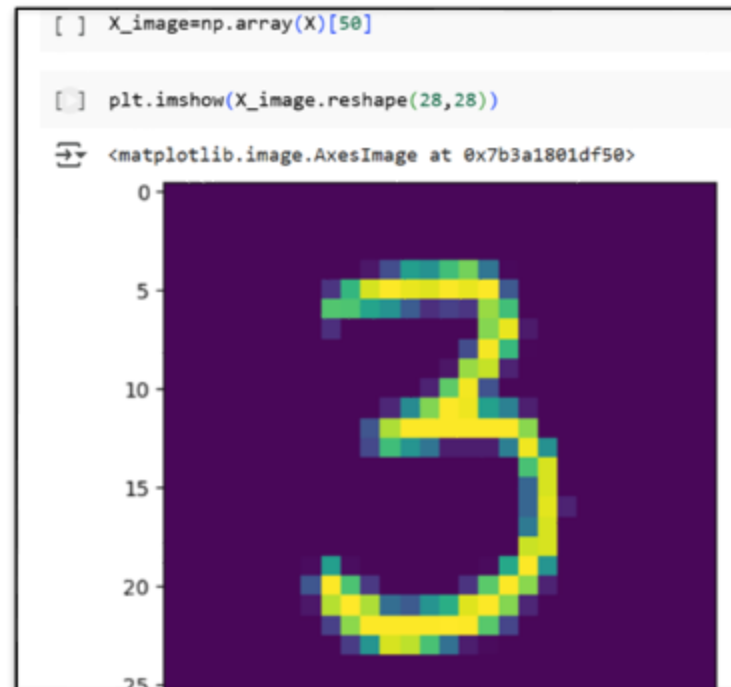
```
>>> Missing values: 0
```

VISUALIZATIONS

MNIST Sample Visualization :
Shows first 5 labeled MNIST digits.

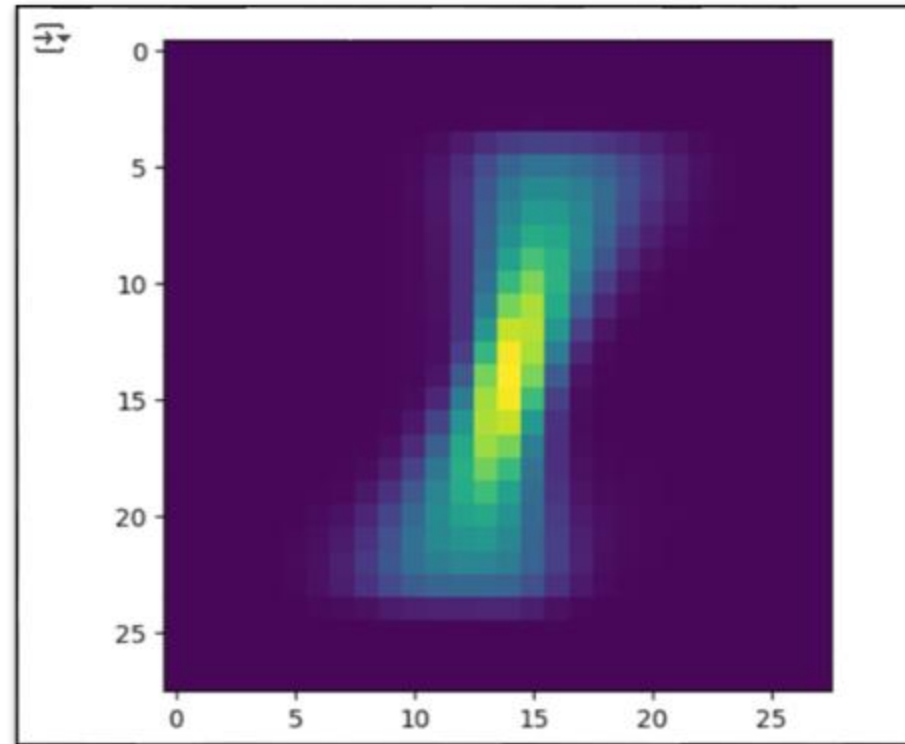


Displaying a Single MNIST Image:
Shows the 51st MNIST image as a 28x28 pixel digit ("3").

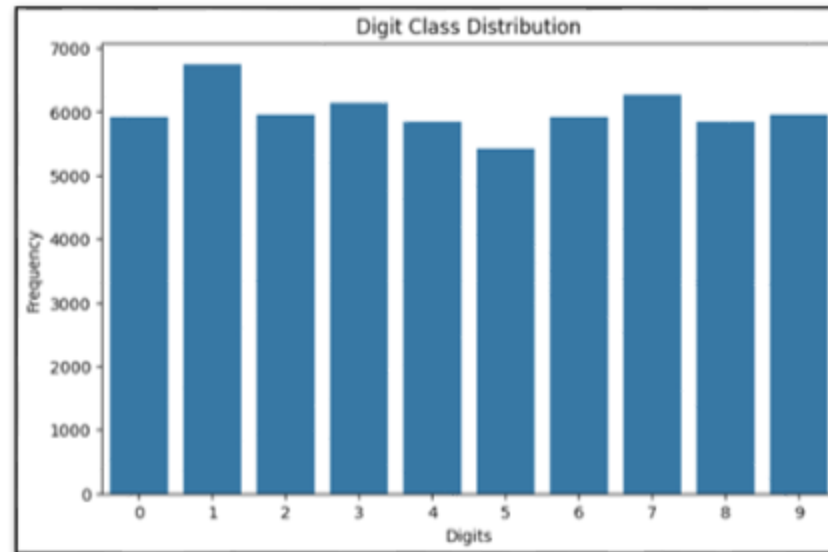


Average Digit Visualization:

Calculates and displays the average image of digit '1' from the MNIST dataset.



MNIST Digit Distribution: Bar chart showing the frequency of each digit (0-9) in the MNIST dataset.



STREAMLINED GRADING PROCESS

- 1. Image Capture:** Scan or photograph the student's worksheet.
- 2. Digit Recognition:** Use the trained ML model to identify handwritten digits.
- 3. Answer Verification:** Compare recognized digits with correct answers.
- 4.Score Calculation:** Assign points based on accuracy and clarity of handwriting as full, partial, or zero marks
- 5. Report Generation:** Produce a concise score summary for the student and send results to LMS (Google Classroom, Blackboard)

MACHINE LEARNING MODEL FOR DIGIT RECOGNITION

- The next step after data preparation was to train a machine learning model for predictive analysis.
- For MNIST digit recognition, Gaussian Naive Bayes outperformed standard Naive Bayes, achieving 94.73% accuracy on test data after parameter tuning and feature normalization

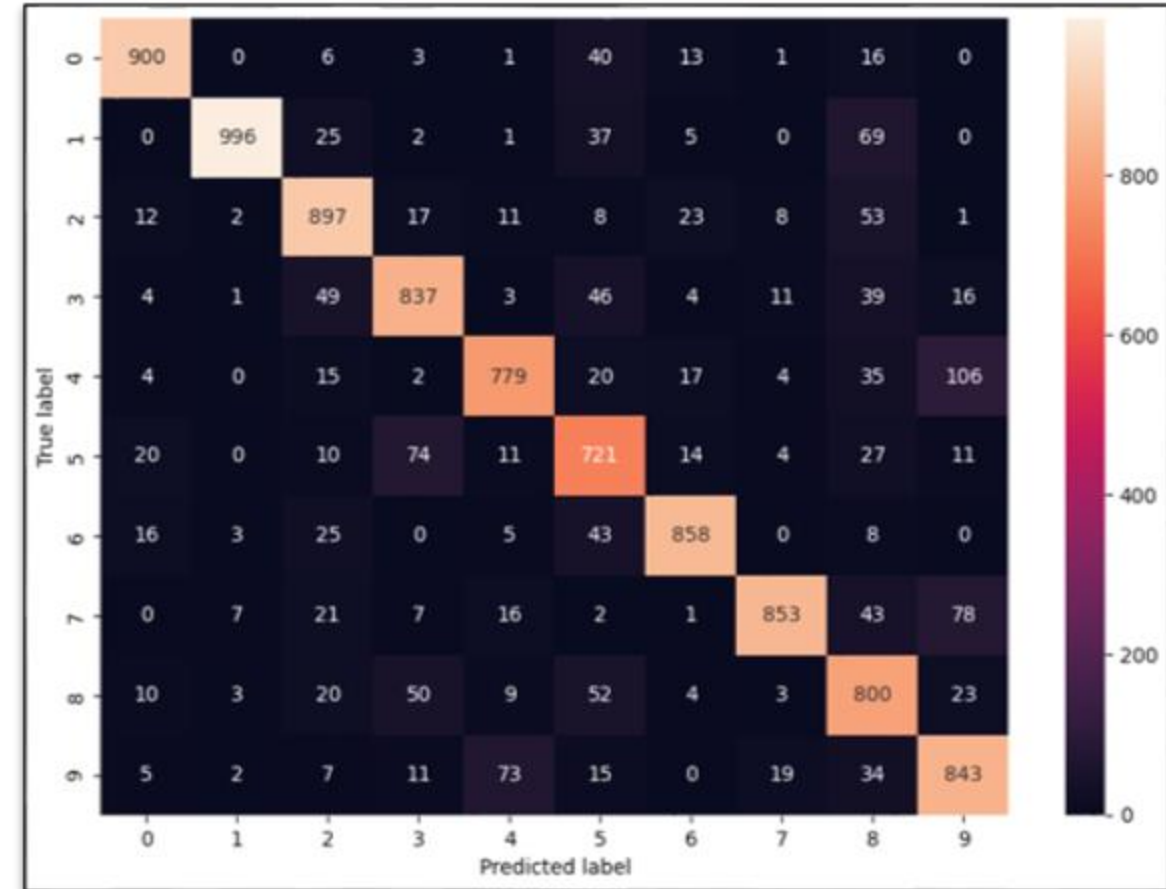


NAÏVE BAYES CLASSIFIER

- This classification model uses Bayes' theorem for probabilistic predictions. $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$
- Advantage:**
- Simple and fast, good for quick baseline results.
- Disadvantage:**
- Assumes pixels are independent, which isn't true for images, limiting accuracy.
- With 84% accuracy, below our threshold, this model was rejected.

```
[ ] # for testing data
y_hat_test=gnb.predict(X_test)
accuracy(y_test,y_hat_test)
```

```
np.float64(0.8484)
```

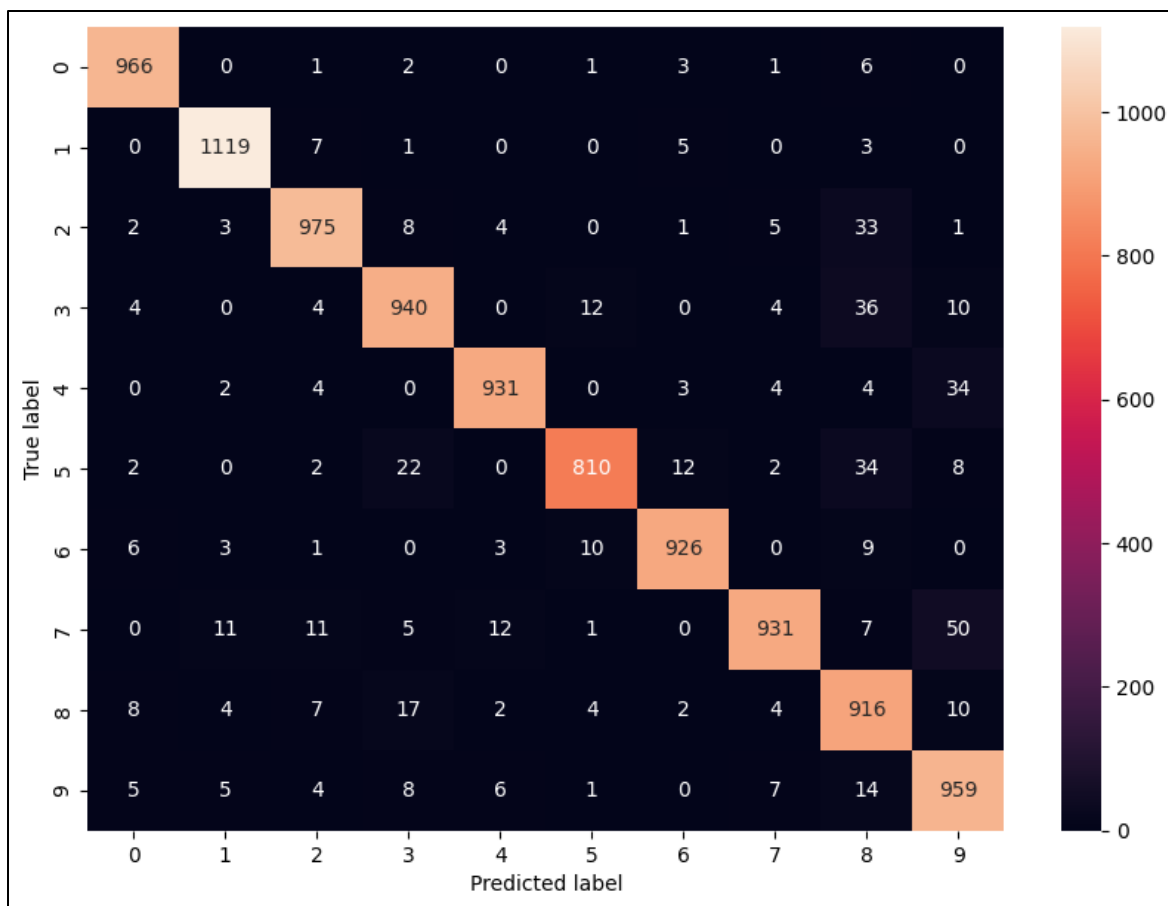


NON-NAÏVE GAUSS BAYES CLASSIFIER

- Gaussian Bayes, unlike Naive Bayes, considers pixel correlations in images, leading to higher (94%) accuracy on MNIST's complex digit patterns.
- I normalized **X_test** by dividing by 255 and fine tuning the epsilon our hyper-parameter to improve results.

```
[ ] # Non-naive bayes for testing data
y_hat_test2=gaussb.predict(X_test)
accuracy(y_test,y_hat_test2)
```

```
np.float64(0.9473)
```



NEXT PROCESS IN AUTOMATED GRADING

- **Answer Comparison & Grading:** Recognized digits are matched against an answer key, assigning scores based on accuracy.
- **Score Report Generation:** Detailed score reports with feedback are created.
- **LMS Integration:** Reports are integrated into LMS platforms (e.g., Google Classroom, Blackboard) for streamlined feedback.

Aspect	Criteria	Points Awarded
Correct	Recognized digits matches the expected answer	1 point
Minor variation	Slightly unclear but resembles correct digit	0.5 point
Wrong	Recognized digit is incorrect	0 point
Illegible	System cannot confidently determine digit	Manual review

Question	Expected Answer	Recognized Answer	Score
3+5	8	8	1
7+2	9	6	0
2+9	11	11(messy)	0.5
4+3	7	7	1
Tota Score =2.5/4 = (62.5%)			

KEY BENEFITS OF AUTOMATED GRADING

Time-Saving & Efficiency – Reduces manual grading workload, allowing teachers to focus on personalized instruction.

Consistent & Fair Evaluation – Eliminates human bias, ensuring uniform assessment for all students.

Instant Feedback – Provides real-time results, helping students learn from mistakes immediately.

Seamless LMS Integration – Connects with Google Classroom, Blackboard, and other platforms for streamlined reporting.

Encourages Student Engagement – Motivates students with immediate performance insights and targeted feedback.

CONCLUSION AND FUTURE ENHANCEMENTS

- The system automates grading with 94% accuracy using Gaussian Bayes, outperforming the Naive Bayes baseline model.
- For future enhancements we can enhance accuracy with CNNs, support multi-digit input, provide AI handwriting feedback, and integrate with online learning platforms.





APPENDIX

*All images from google search

*Source of dataset (Modified National
Institute of Standards and Technology database) :
Kaggle

A series of thin, white, overlapping geometric lines and polygons on the left side of the slide, creating a modern, abstract design.

THANK YOU